Stability of ground vehicles.

1 Introduction

In this lesson, we introduce examples of stability analysis for road and rail vehicles. The most important field force acting on a road vehicle consists of the force applied at the contact between the wheel and its support (either road or rail). We devote the first part of this lesson (cf Section 2) to introducing some concepts about the forces generated at the wheel-road interface. To this aim, we start from the simple case of a rigid disk rolling on a rigid support and refine this model introducing the effect of elastic flexibility in the contacting bodies.
Then, we introduce a simplified model of a road vehicle running through a curve and use that model to investigate the possible occurrence of divergence-type instability, depending on the vehicle's design parameters and speed. Finally, we introduce a simple model of a railway vehicle (actually, of a single wheelset constrained by primary suspensions) and use this model to introduce hunting instability, a flutter-type dynamic instability that may occur in railway vehicles at relatively large speed

2 Elements of rolling contact theory.

2.1 Rolling contact of rigid bodies, the Coulomb's theory

First of all, we recall the simple theory dealing with the case of a rigid disk rolling on a rigid support, as shown in Figure 2.1. Because the contacting bodies are rigid, the contact region reduces to point P. If we assume no slip in the contact point, then P is the disk's centre of rotation and the speed of the disk centre C is:

$$v_c = \omega R \tag{2.1}$$

with ω the angular speed and R the radius of the disk.
According to the Coulomb's theory of dry friction, to avoid slip in the contact point the normal and tangential contact force components in the contact point, N and T respectively, need to satisfy the following disequation:

$$\frac{|T|}{N} \leq f_s \tag{2.2}$$

with f_s the static friction coefficient for the contact between the two bodies.

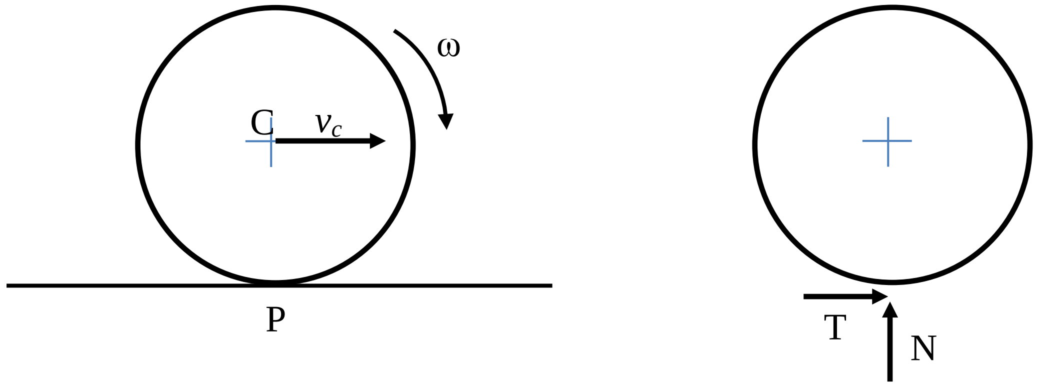


Figure 2.1: Rigid disk rolling on a rigid straight support (left). Forces applied by the support on the disk (right)

If otherwise slip occurs between the two bodies, there is no kinematic relationship between the angular speed and the speed of the centre of the disk, whereas the modulus of the tangential force is related to the normal force as:

$$|T| = f_d N \tag{2.3}$$

with $f_d \le f_s$ the dynamic friction coefficient.

2.2 *Elements of the theory of rolling contact between deformable elastic bodies – in-plane motion*

If we assume the two contacting bodies to be deformable, then a finite contact region (called contact patch) takes place. However, there are deeper consequences of assuming the bodies to be deformable. In particular, we will show in this section that:

i) the speed of the disk centre C is normally different from the 'kinematic' value expressed by (2.1), also in case (2.2) is satisfied;

ii) There is a relationship between the contact force and the kinematic quantities defining the movement of the disk over the support, in other words, the contact force constitutes a force field that under certain circumstances can affect the stability of a ground vehicle;

iii) the contact patch is divided in a leading portion where adhesion of the wheel on the road takes place and a trailing portion where slip occurs.

We assume here the wheel's material to have linear elastic behaviour, and we assume the deformation of the support to be negligible. This latter assumption is well justified for a road vehicle, since the tyres are much more deformable than the road. Anyway, no loss of generality is implied by this assumption, because infinitesimal deformations have the same effect when applied at any of the two contacting bodies.

Based on the assumptions made, the contact patch is a segment whose length we denote by $2a$, cf. Figure 2.2. We introduce a coordinate ξ running in the contact patch with zero at the leading edge of the patch. Within the contact segment, the support and the disk exchange a distribution of normal forces per unit length $p(\xi)$ and a distribution of tangential forces per unit length $\tau(\xi)$. The normal and tangential contact forces N and T are the resultant of the normal and tangential forces per unit length respectively:

$$N = \int_0^{2a} p(\xi)d\xi \;\; ; \;\; T = \int_0^{2a} \tau(\xi)d\xi \tag{2.4}$$

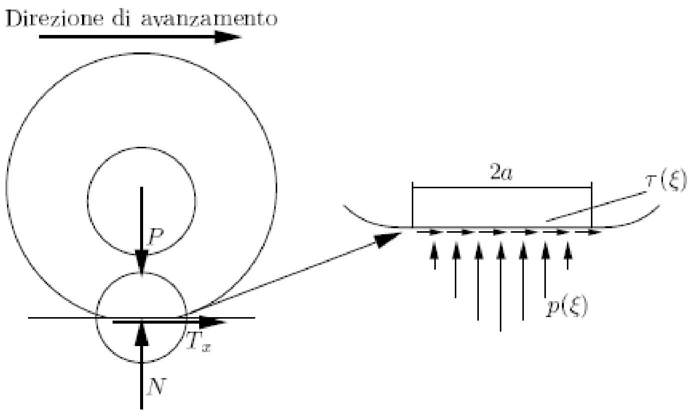


Figure 2.2: Deformable disk rolling on a rigid straight support. Left: general view. Right: detail of the contact patch showing the forces per unit length in normal and tangential direction.

It can be shown [1] that under rather general assumptions the tangential forces per unit length generated in the contact patch are not affecting the size of the contact patch and the distribution of the normal forces per unit length. Hence, we first define the distribution of the normal contact forces, then we use this to determine the tangential forces.
Based on the Hertz's theory, the distribution of normal pressures can be approximated as a parabola. Formulae are available that define the size of the contact patch $2a$ and the maximum normal force per unit length as function of the resulting normal force N, the curvature of the wheel and the elasticity properties of the materials. These formulae are however of minor interest here, and are not reported.
The tangential forces per unit length result from the tangential deformation of the disk in the contact region, which in turn depends on the movement of the disk, on the distribution of normal forces per unit length and on the frictional properties of the contact. To show this we use a simple model known as the *brush model*.
The brush model assumes that each infinitesimal element of the contact patch deforms tangentially in the same way as the single pin of a brush, i.e. the local deformation $u(\xi)$, cf. Figure 2.3, only depends on the local force applied in tangential direction. Note that in reality the deformation of the wheel at any given location inside the contact patch depends on the tangential forces applied across the whole contact region. In order to consider this situation however, more complicated (and accurate) contact models should be used, that are not within the aims of this lesson.

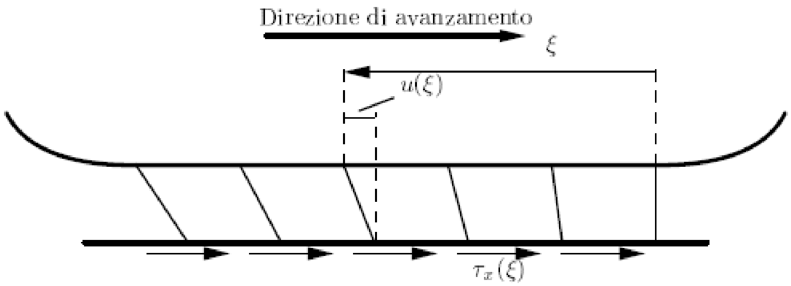


Figure 2.3: Tangential deformation of the wheel in the contact patch.

Owing to the assumption of linear material behaviour, the deformation is described as proportional to the force per unit length according to:

$$\tau(\xi) = Cu(\xi) \tag{2.5}$$

with C the stiffness per unit length (in tangential direction) of the contact.
Now we consider one single pin of the brush flowing through the contact region while the wheel rolls. The speed of the lower end of the pin $v(\xi)$ is:

$$v(\xi) = v_c - \omega R + \frac{du}{dt} = v_c - \omega R + \frac{du}{d\xi}\frac{d\xi}{dt} = v_c - \omega R + v_c \frac{du}{d\xi} \tag{2.6}$$

with v_c the speed of the wheel centre, ω the wheel's angular speed.
Wherever possible, the lower end of the pin tends to adhere to the road: when this happens, the speed expressed by (2.6) reduces to zero:

$$v(\xi) = v_c - \omega R + v_c \frac{du}{d\xi} = 0 \tag{2.7}$$

Solving (2.7) for the gradient of the tangential deformation u provides

$$\frac{du}{d\xi} = -\frac{v_c - \omega R}{v_c} \tag{2.8}$$

Based on (2.8), we define the longitudinal creepage ε_l of the wheel moving over the support the following non-dimensional ratio:

$$\varepsilon_l = \frac{v_c - \omega R}{v_c} \tag{2.9}$$

Based on (2.5), (2.8) and (2.9) we have:

$$\frac{d\tau}{d\xi} = -C\frac{du}{d\xi} = -C\frac{v_c - \omega R}{v_c} = -C\varepsilon_l \tag{2.10}$$

Finally, we observe that the brush pin enters the contact patch with zero deformation, and therefore the tangential force per unit length must reduce to zero in this point:

$$\tau(0) = 0 \tag{2.11}$$

so that altogether (2.10) and (2.11) provide:

$$\tau(\xi) = -C\varepsilon_l\xi \tag{2.12}$$

Based on (2.12) the tangential force per unit length grows linearly with the distance from the leading edge of the contact patch, with proportionality coefficient depending on the longitudinal creepage ε_l and on the tangential contact stiffness per unit length C.
We recall that this result was obtained under the assumption that the lower tip of the brush adheres on the road. If we apply Coulomb's law of dry friction to each pin in the brush, we conclude that we have adhesion in the contact region where the following inequality holds:

$$|\tau(\xi)| \leq f_s p(\xi) \tag{2.13}$$

Figure 2.4 (left) compares the absolute value of the tangential force per unit length τ according to (2.12) (dashed straight line) with the *adhesion limit*, i.e. the normal force per unit length p multiplied by the static friction coefficient f_s (dashed parabola). The expression of $\tau(\xi)$ according to (2.12) is valid only when (2.13) holds, i.e. when the straight line is below the parabola. In this region, there is adhesion between the wheel surface (i.e. the tips of the brush) and the road. In the remaining portion of the contact patch, adhesion of the two surfaces is not possible, and the tangential force per unit length is saturated to $f_d\,p(\xi)$ with f_d the dynamic friction coefficient, being $f_d \leq f_s$. The resulting distribution of the tangential forces per unit length is shown in Figure 2.4 (right) for the case $f_d < f_s$. An increase in the longitudinal creepage ε_l results in slip taking place in a larger portion of the contact patch.

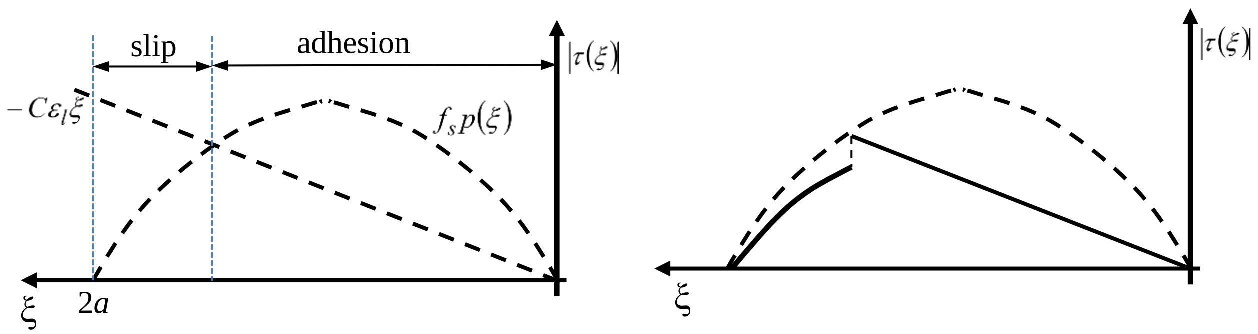


Figure 2.4: Left: adhesion and slip regions in the contact patch. Right: trend of the tangential foprce per unit length in the contact patch.

The magnitude of the resulting tangential force T, cf. (2.4) is the area below the curve representing $\tau(\xi)$ in Figure 2.2 right. If we assume $f_d = f_s$, this area is monotonically increasing with the longitudinal creepage, resulting in a diagram of $T(\varepsilon_l)$ as represented by the solid line in Figure 2.5. If otherwise $f_d < f_s$ the trend of $T(\varepsilon_l)$ is initially increasing but at large creepage values becomes decreasing on account of the actual tangential force being lower than the adhesion limit on the slip region, as shown by the dashed line in Figure 2.5. We see that, as anticipated above in this section, there is a functional relationship between the magnitude of the tangential force T and the longitudinal creepage, which ultimately depends on the kinematics of the wheel, so the contact force is state-dependent or, in other words, a force field. We also note that, except the special case T=0, the longitudinal creepage is not zero, hence the speed of the disk centre is not the 'kinematic value' expressed by (2.1).

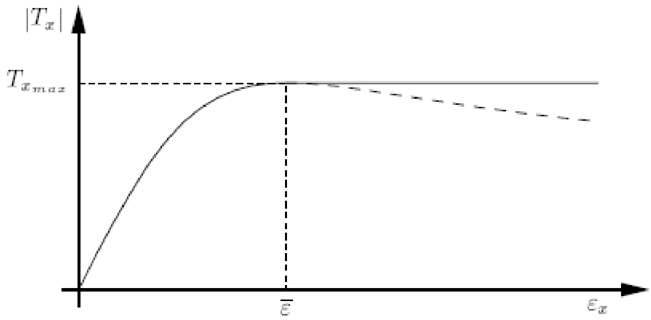


Figure 2.5: Trend of the tangential force T with the longitudinal creepage ε_l. Solid line: $f_d = f_s$. Dashed line: $f_d < f_s$.

2.3 *Rolling contact between deformable elastic bodies – generalisation to the out-of-plane motion*

The model presented in Section 2.2 for the rolling contact between elastic bodies can be generalised to also consider 'out of plane' motion components, i.e. a general motion of the wheel over the support.

First of all, we note that in a generic three-dimensional motion the speed of the centre of the wheel is not necessarily lying in the plane of the wheel. Hence, we generalise the definition of creepage (2.9) defining the creepage vector $\vec{\varepsilon}$ as follows:

$$\vec{\varepsilon} = \frac{\vec{v}_c - \vec{\omega} \wedge \vec{R}}{|\vec{v}_c|} \tag{2.14}$$

with $\vec{v}_c$ the vector denoting the speed of the wheel centre, $\vec{\omega}$ the wheel angular velocity vector, $\vec{R}$ the position of the wheel centre with respect to the 'nominal' contact point (i.e. the centre of the contact patch, neglecting the elastic deformation).

The creepage vector can be decomposed into longitudinal and lateral components:

$$\vec{\varepsilon} = \varepsilon_l \vec{l} + \varepsilon_t \vec{t} \tag{2.15}$$

with $\vec{l}$ and $\vec{t}$ the unit vectors pointing in the longitudinal and transversal direction. The longitudinal direction is the one defined as the intersection of the contact plane with the wheel plane, the transversal direction is parallel to the contact plane and normal to the longitudinal direction. The longitudinal and transversal creepage components are defined as:

$$\varepsilon_l = \frac{v_{cl} - \omega R}{|\vec{v}_c|}$$
$$\varepsilon_t = \frac{v_{ct}}{|\vec{v}_c|} \tag{2.16}$$

Assuming small creepages (i.e. the wheel is not in full slip condition), the difference between the longitudinal component and the magnitude of the wheel centre speed v_{cl} and $|\vec{v}_c|$ is negligible, and the first of (2.16) coincides with (2.9), whereas the second of (2.16) can be interpreted as the tangent of the angle α formed by vector $\vec{v}_c$ with the wheel plane, cf. Figure 2.6.
Because for small angles:

$$\arctan(\alpha) \cong \alpha$$

the transversal creepage ε_t can be also defined as angle α which in road vehicle dynamics is called the *slip angle* (or *sideslip angle*).

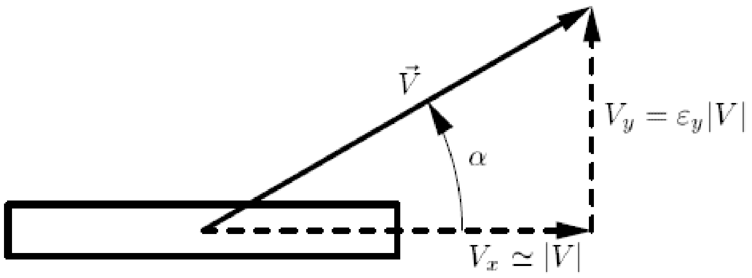


Figure 2.6: Geometrical interpretation of the transversal creepage or slip angle.

The force exchanged by the wheel with the support (road or rail) is composed by a normal component N perpendicular to the contact plane and by a tangential component, laying in the contact plane. The tangential component can be further decomposed into a longitudinal component T_l parallel to $\vec{l}$ and a transversal component T_t, directed along $\vec{t}$. The tangential components of the contact force are often called *creepage forces*.
On account of the finite size of the contact region, the wheel is also subject to a moment M generated by the contact forces along an axis perpendicular to the contact plane and passing through the centre of the contact patch and called *spin moment*. As far as rail vehicles are concerned, the effect of the spin moment on vehicle dynamics is usually very small due to the small size of the contact patch and therefore this moment is usually neglected. For road vehicles the spin moment is responsible for the self-aligning effect in steered wheels, and therefore cannot be neglected when evaluating the forces needed to steer the vehicle, although still it is sometimes neglected when studying the dynamics of the whole vehicle.
The creep forces can be shown to depend upon the normal contact force component N and the creepage components ε_l and ε_t:

$$T_l = T_l(\varepsilon_l, \varepsilon_t, N)$$
$$T_t = T_t(\varepsilon_l, \varepsilon_t, N) \tag{2.17}$$

Often, this dependence is expressed in the form:

$$T_l = -\mu_l(\varepsilon_l, \varepsilon_t, N)N$$
$$T_t = -\mu_t(\varepsilon_l, \varepsilon_t, N)N \tag{2.17b}$$

with μ_l and μ_t defined as the generalised friction coefficients for the longitudinal and lateral direction.

3 Dynamics of a road vehicle in a curve – the basics.

In this section, we introduce a very simple model of a road vehicle and use it to introduce the basic features of the behaviour of a road vehicle in a curve. Referring to a standard four-wheeled car with steering wheels at the front, we introduce the model shown in Figure 3.1, often called the *bicycle model* as it reduces the two front wheels to one single steered wheel (with δ the steer angle) and the two rear wheels to a single non-steered wheel.
This model drastically simplifies the kinematics of the steering system and neglects some important effects such as the load transfer from the inner to the outer wheels and the movements of the wheels relative to the car body allowed by compliance in the vehicle suspension, but it is sufficient for the scopes of this lesson.
In Figure 3.1 δ is the steer angle, T_{lf} and T_{tf} the longitudinal and transversal creep forces on the front axle, T_{lr} and T_{tr} the same forces on the rear axle.

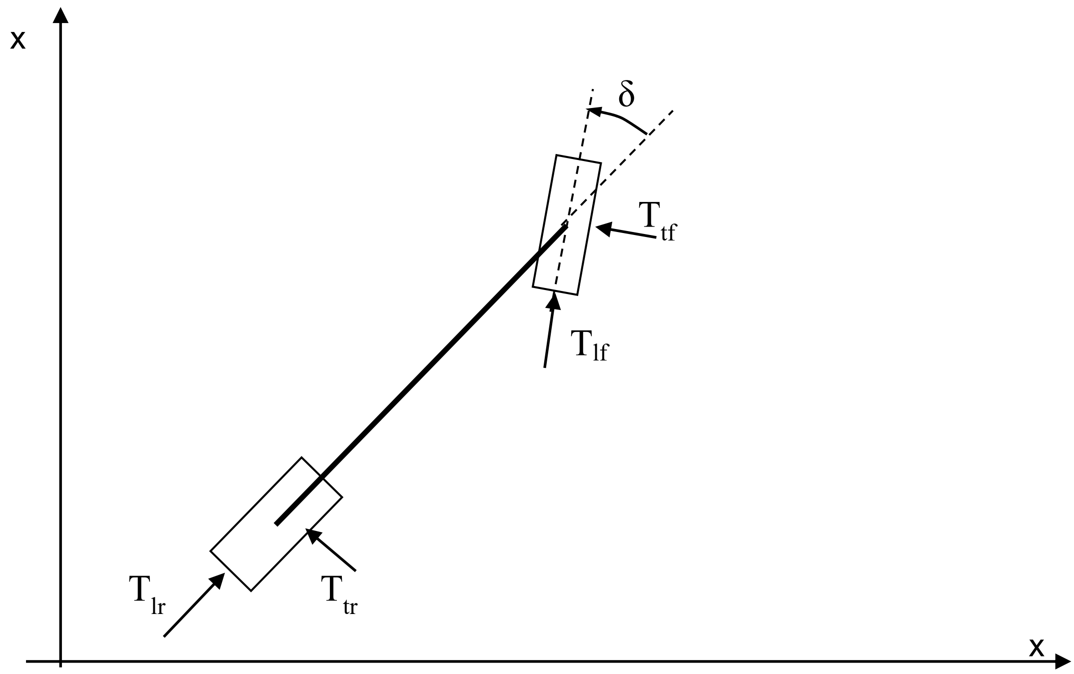


Figure 3.1: The bicycle model for the study of road vehicle dynamics in a curve.

We perform our study under the following simplifying assumptions:

i) Small steering angle (i.e. large radius of the curve);
ii) small slip angles (i.e. wheels are far from full slip condition)
iii) the car moves with constant forward speed *V*;
iv) traction and braking forces are neglected, and therefore the longitudinal creep forces on the front and rear axle are negligible;
v) spin moments at the wheels are neglected;
vi) aerodynamic forces are neglected.

3.1 Steady-state curving

We consider first a condition when a constant steering angle δ is applied and we look at the motion of the vehicle after the transient produced by the application of the steering angle becomes negligible and the vehicle achieves the so-called steady-state condition. Note the above statement assumes for the time being the vehicle to show an asymptotically stable behaviour. We will actually analyse the stability of the vehicle in Section 3.2.

First, we consider an ideal condition in which no slip occurs on the front and rear axles. In this case, the speed of the front and rear wheels, $\vec{V}_f$ and $\vec{V}_r$ respectively, are parallel to the centre plane of the two wheels. Therefore, the angle φ formed by the lines connecting the centre of rotation of the car body (point C in the figure) with the centre of the front and rear wheels equals the steer angle δ. Now, we consider the circumference centred in C and passing through the centres of the front and rear wheels. This is the constant radius curve being negotiated by the vehicle and we denote as *R* the radius of the curve. For small steer angle, angle φ is also small and is well approximated by the ratio of the chord *L* (the vehicle wheelbase) over the radius of the curve *R*, hence:

$$\delta = \varphi = \frac{L}{R} \tag{3.1}$$

The above equation provides a relationship between the steering angle and the radius of the curve negotiated by the vehicle. under the no-slip assumption, this relationship is not influenced by the vehicle speed, which is not in agreement with our experience as drivers.

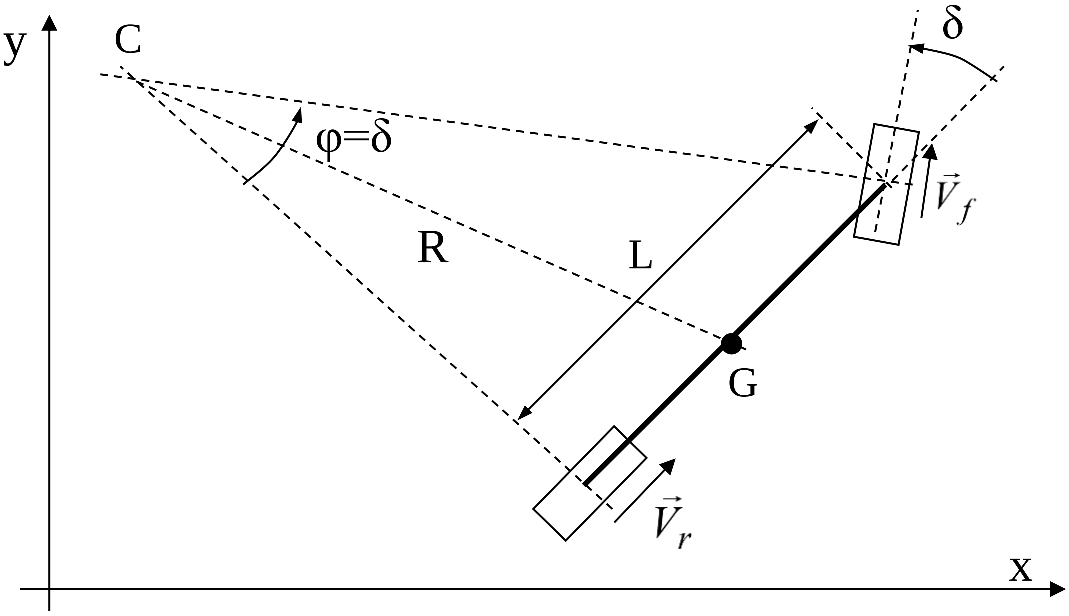


Figure 3.2: Kinematics of the bicycle model for zero slip angle on the front and rear axles

If we consider instead the effect of slip angles actually taking place at the front and rear wheels, we have the situation displayed in Figure 3.3. In this case, the following relationship holds:

$$\delta - \alpha_f + \alpha_r = \varphi = \frac{L}{R}$$

which can be solved for δ bringing:

$$\delta = \frac{L}{R} + \alpha_f - \alpha_r \tag{3.2}$$

For zero longitudinal creep force and for constant component of the contact force, the slip angles on the front and rear wheels are uniquely related to the transversal contact forces T_{tf} and T_{tr} taking place on the two wheels:

$$\begin{aligned} T_{tf} &= T_{tf}(\alpha_f) \\ T_{tr} &= T_{tr}(\alpha_r) \end{aligned} \tag{3.3}$$

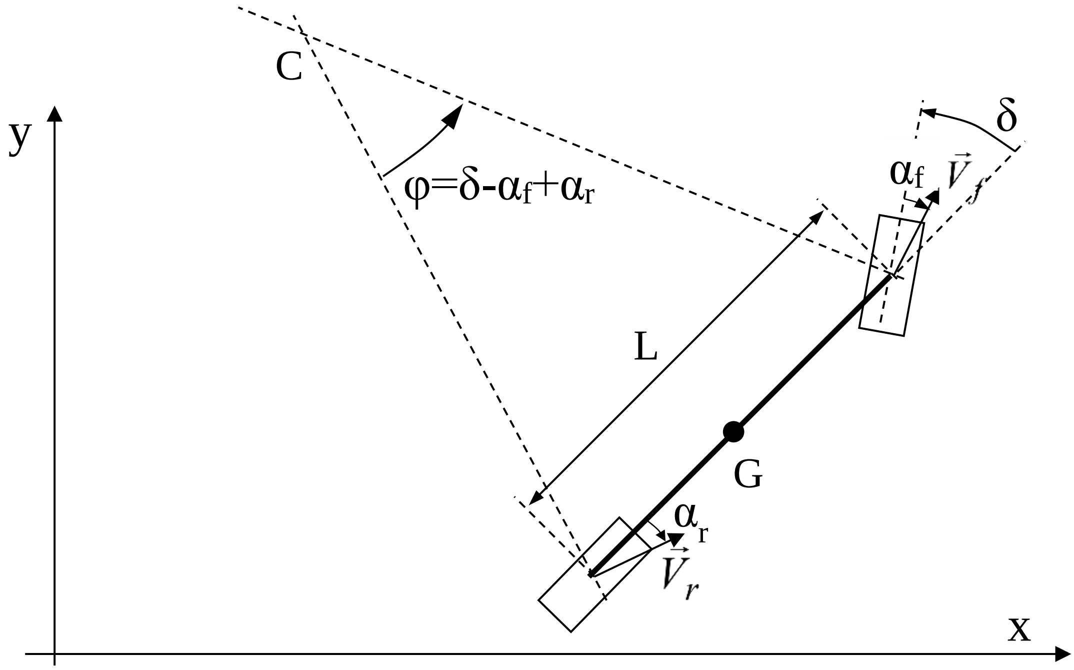


Figure 3.3: Kinematics of the bicycle model for zero slip angle on the front and rear axles

Figure 3.4 shows the qualitative trend of the transversal contact force with the slip angle for a road tire: it is observed that for small slip angles, the non-linear relationships (3.3) are well approximated by the following linear relationships:

$$\begin{aligned} T_{tf} &= C_f \alpha_f \\ T_{tr} &= C_r \alpha_r \end{aligned} \tag{3.4}$$

In (3.4) C_f and C_r are the gradients of the transversal force with slip, and are named *the cornering stiffness* of the front and rear axles (note that from the point of view of dimensional analysis these parameters actually are forces and not forces per unit length).

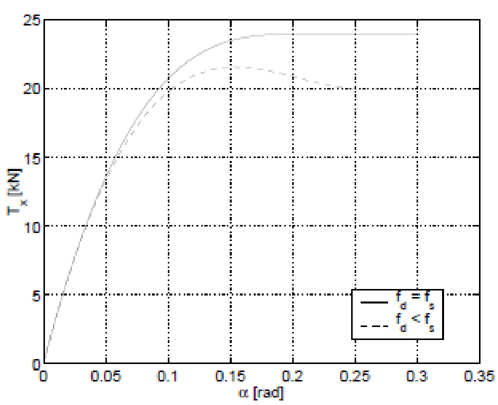


Figure 3.4: Trend of the transversal creep force with the slip angle for a road tire.

Figure 3.5 shows the forces acting on the vehicle i.e., under assumptions i) to vi) , the transversal contact forces and the centrifugal force acting on the car body centre of mass. The forces T_{tf} and T_{tr} are obtained from the static equilibrium of the vehicle:

$$\begin{aligned} &T_{tf}\cos(\delta) + T_{tr} = m\frac{V^2}{R} \\ &T_{tf}\cos(\delta)a - T_{tr}b = 0 \end{aligned} \tag{3.5}$$

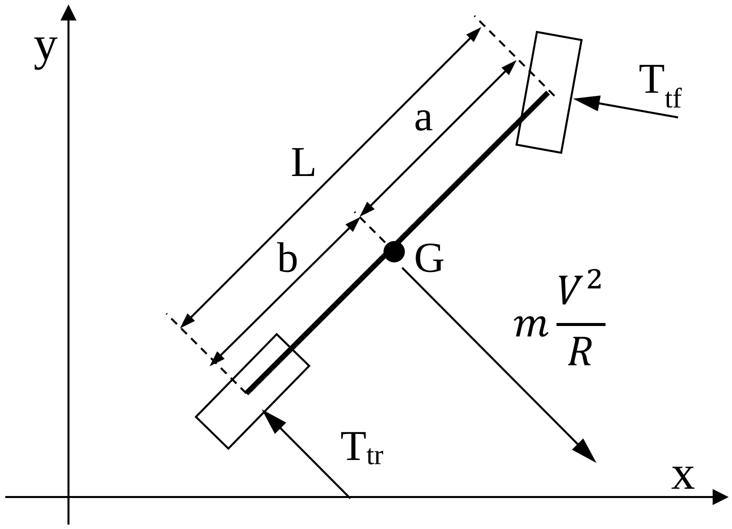


Figure 3.5: Kinematics of the bicycle model for zero slip angle on the front and rear axles

For small steer angle

$\cos(\delta) \cong 1$

and, solving (3.5) for T_{tf} and T_{tr} brings:

$$\begin{aligned} &T_{tf} = m\frac{V^2}{R}\frac{b}{L} \\ &T_{tr} = m\frac{V^2}{R}\frac{a}{L} \end{aligned} \tag{3.6}$$

replacing (3.4) and (3.6) in (3.2) brings.

$$\delta = \frac{L}{R} + m\frac{V^2}{RL}\left(\frac{b}{C_f} - \frac{a}{C_r}\right) \tag{3.7}$$

We define the vehicle's understeer coefficient K as:

$$K = \frac{m}{L^2}\left(\frac{b}{C_f} - \frac{a}{C_r}\right) \tag{3.8}$$

so that equation (3.7) is rewritten as:

$$\delta = \frac{L}{R}\left(1 + KV^2\right) \tag{3.9}$$

Equation (3.9) shows that the steer angle required to negotiate a curve with constant radius R is actually affected by the vehicle's speed. If the understeer coefficient K is greater than zero, then the

steer angle is increasing with the square of the speed: this is called an *under-steering behaviour* of the vehicle. If otherwise $K < 0$ the steer angle is decreasing with speed, this is called an *over-steering behaviour* of the vehicle. In the particular case $K = 0$ the steering angle is not affected by the vehicle speed: this is called a *neutral behaviour* of the vehicle.
Figure 3.6 shows the trend of the steering angle with speed for under-steering, neutral and over-steering vehicle behaviour. In this latter case, the steer angle reduces to zero at a certain speed $\overline{V}$ and theoretically should be lower than zero for higher speeds. It will be shown in the next section that at speed $\overline{V}$ the over-steering vehicle becomes unstable with a divergence type instability.

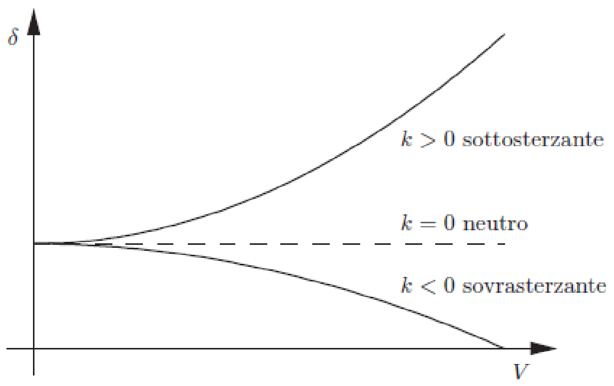


Figure 3.6: Trend of the steering angle δ with speed for an understeering, neutral and oversteering vehicle.

Before closing this section, we briefly mention some design parameters that are affecting the under-steering / over-steering behaviour of the vehicle. Owing to the equation (3.8) defining the understeer coefficient:

i) moving the vehicle's centre of mass towards the front axle will reduce semi-wheelbase *a* and increase semi-wheelbase *b* and hence will increase the tendency to understeering;
ii) Decreasing the sideslip stiffness C_f of the front axle and / or decreasing the sideslip stiffness C_r of the rear axle will increase the tendency to understeering. The value of the sideslip angle can be affected by introducing longitudinal creepages on the two axles (e.g. through braking or traction) and / or by changing the properties of the tires (mostly materials and shape), or by increasing / reducing load transfer e.g. by the use of torsion bars (an effect that is not directly captured by the model used in this section).

3.2 Stability analysis

In this section we analyse the stability of the steady-state steering condition for the road vehicle as defined in Section 3.1. Considering again the two wheel model, the road vehicle can be modelled as a 3-d.o.f. system having as independent coordinates the two displacement components of the car body centre of mass (c.o.m.) G and the car body yaw rotation and assuming the steer angle δ to be a known input of the system.

It is possible however to reduce the number of state variables involved in the problem. Indeed, we note that the forces acting on the vehicle are not affected by the position of the vehicle, but only by the velocity. Furthermore, the vehicle is assumed to move at constant speed, so that the velocity of the c.o.m. is fully defined by a single state variable, the car body sideslip angle β defined as the angle formed by the velocity vector with the car body centre plane, cf. Figure 3.7. The second state variable needed to describe the dynamic behaviour of the vehicle is the yaw angular speed $\dot{\psi}$, also show in the same figure.

In order to write the equations of motion for the vehicle, we first define the expression of the inertia forces. Referring again to Figure 3.7 and identifying the *x* and *y* axes with the real and imaginary axes of the complex plane, the speed of the c.o.m. can be written as a complex number in polar form:

$$\vec{V}_G = Ve^{i(\beta+\psi)} \tag{3.10}$$

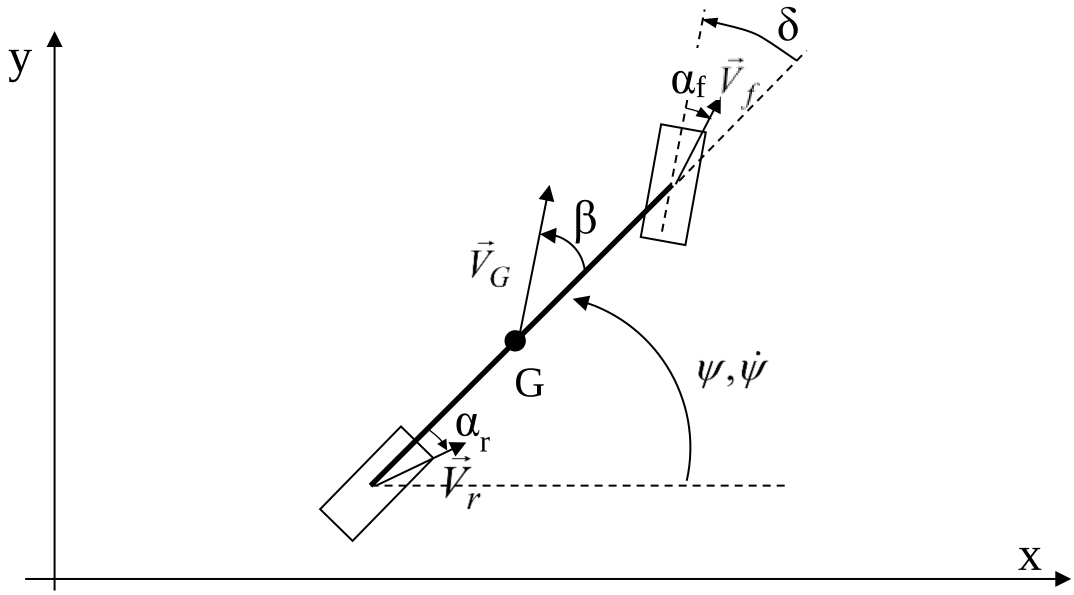


Figure 3.7: Kinematics of the two-wheel car model.

The car body acceleration is:

$$\vec{a}_G = \frac{d\vec{V}_G}{dt} = i(\dot{\beta}+\dot{\psi})Ve^{i(\beta+\psi)} = -(\dot{\beta}+\dot{\psi})V\sin(\beta)e^{i\psi} + i(\dot{\beta}+\dot{\psi})V\cos(\beta)e^{i\psi} \tag{3.11}$$

and, neglecting 2nd order and higher terms:

$$\vec{a}_G = i(\dot{\beta}+\dot{\psi})Ve^{i\psi} \tag{3.12}$$

Equation (3.12) can be used to define the inertia force on the vehicle, which is directed orthogonal to the car body mid-plane. Figure 3.8 shows all the forces and moments acting on the vehicle, according to assumptions i)-vi) defined above.

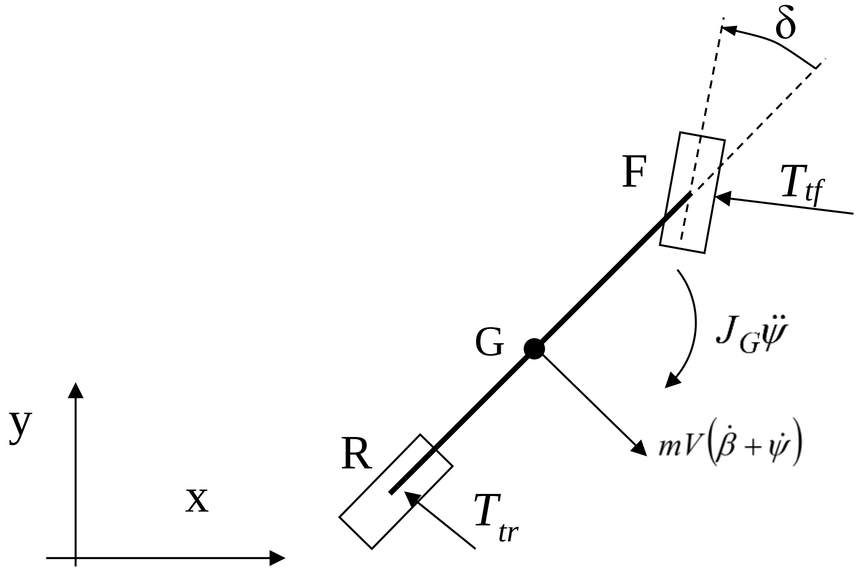


Figure 3.7: Kinematics of the two-wheel car model.

The dynamic equilibrium of the vehicle provides the two equations:

$$\begin{aligned} &T_{tf}\cos(\delta)+T_{tr}=mV\left(\dot{\beta}+\dot{\psi}\right)\\ &T_{tf}\cos(\delta)a-T_{tr}b=J_G\ddot{\psi}\end{aligned} \qquad (3.13)$$

For small steer angle δ (3.13) can be simplified neglecting 2dn order terms:

$$\begin{aligned} &T_{tf}+T_{tr}=mV\left(\dot{\beta}+\dot{\psi}\right)\\ &T_{tf}a-T_{tr}b=J_G\ddot{\psi}\end{aligned} \qquad (3.14)$$

Next, we define the contact forces T_{tf} and T_{tr} as function of the state variables and their derivatives. To this aim, we define first a mathematical expression for the speed of the front and rear wheels:

$$\begin{aligned} &\vec{V}_F=\vec{V}_G+\vec{V}_{FG}=Ve^{i(\beta+\psi)}+a\dot{\psi}e^{i\left(\psi+\frac{\pi}{2}\right)}=V\cos(\beta)e^{i\psi}+(V\sin(\beta)+a\dot{\psi})e^{i\left(\psi+\frac{\pi}{2}\right)}\\ &\vec{V}_R=\vec{V}_G+\vec{V}_{RG}=Ve^{i(\beta+\psi)}-b\dot{\psi}e^{i\left(\psi+\frac{\pi}{2}\right)}=V\cos(\beta)e^{i\psi}+(V\sin(\beta)-b\dot{\psi})e^{i\left(\psi+\frac{\pi}{2}\right)}\end{aligned} \qquad (3.15)$$

Again neglecting non-linear terms in (3.15) a simplified expression is obtained:

$$\begin{aligned} &\vec{V}_F=Ve^{i\psi}+\left(V\beta+a\dot{\psi}\right)e^{i\left(\psi+\frac{\pi}{2}\right)}\\ &\vec{V}_R=Ve^{i\psi}\left(V\beta-b\dot{\psi}\right)e^{i\left(\psi+\frac{\pi}{2}\right)}\end{aligned} \qquad (3.16)$$

For small angles, the following expressions hold for the sideslip angles on the front and rear wheels:

$$\begin{aligned} &\alpha_F=\delta-\frac{V_{F_\perp}}{V_{F_{//}}}\\ &\alpha_R=-\frac{V_{R_\perp}}{V_{R_{//}}}\end{aligned} \qquad (3.17)$$

with $V_{F_\perp}$ and $V_{F_{//}}$ the components of the front wheel velocity vector respectively orthogonal and parallel to the car body and $V_{R_\perp}$ and $V_{R_{//}}$ the same kinematic quantities for the rear wheel. Based on the expression of the velocity vector for the front and rear wheels (3.16), the expressions of the sideslip angles read:

$$\begin{aligned} \alpha_F &= \delta - \frac{V\beta + a\dot{\psi}}{V} = \delta - \beta - \frac{a\dot{\psi}}{V} \\ \alpha_R &= -\frac{V\beta - b\dot{\psi}}{V} = -\beta + \frac{b\dot{\psi}}{V} \end{aligned} \tag{3.18}$$

The contact forces T_{tf} and T_{tr} are then obtained from (3.4) using the expressions of the wheel sideslip angles (3.18). Replacing this in (3.14) brings the following state equations, written in matrix form:

$$[A]\underline{\dot{z}} + [B]\underline{z} = \underline{C}\delta \tag{3.19}$$

with:

$$\underline{z} = \begin{Bmatrix} \beta \\ \dot{\psi} \end{Bmatrix} \; ; \; [A] = \begin{bmatrix} mV & 0 \\ 0 & J_G \end{bmatrix} \; ; \; [B] = \begin{bmatrix} C_F + C_R & mV + \dfrac{C_F a - C_R b}{V} \\ C_F a - C_R b & \dfrac{C_F a^2 + C_R b^2}{V} \end{bmatrix} \; ; \; \underline{C} = \begin{Bmatrix} C_F \\ C_F a \end{Bmatrix} \tag{3.20}$$

The stationary solution of (3.19) for constant steer angle $\delta = \delta_0$ is obtained in the form:

$$\underline{z} = \underline{z}_0 = \text{const} \; ; \; \underline{\dot{z}} = \underline{0} \tag{3.21}$$

Replacing (3.21) in (3.19) brings:

$$\underline{z}_0 = [B]^{-1}\underline{C}\delta_0 \tag{3.22}$$

Noting that $V = R\dot{\psi}$, the same result as per (3.8) and (3.9) is found.
The next step is to analyse the stability of the stationary solution. To this aim, we consider a solution in the form:

$$\underline{z} = \underline{z}_0 + \underline{z}_P \; ; \; \underline{z} = \underline{\dot{z}}_P \tag{3.23}$$

with $\underline{X}_P(t)$ describing the perturbed motion. Replacing (3.23) in (3.19 and recalling (3.22) brings:

$$[A]\underline{\dot{z}}_P + [B]\underline{z}_P = \underline{0} \tag{3.24}$$

The perturbed solution is in the usual form:

$$\underline{z}_P = \underline{Z}e^{\lambda t} + \; ; \; \underline{\dot{z}}_P = \lambda\underline{Z}e^{\lambda t} \tag{3.25}$$

replacing (3.25) in (3.24) we get the linear homogeneous equation:

$$(\lambda[A] + [B])\underline{Z} = \underline{0} \tag{3.26}$$

this equation provides non trivial solutions under the condition:

$$\det(\lambda[A] + [B]) = 0 \tag{3.27}$$

Considering the expressions of matrices [A] and [B] the following 2nd order equation is obtained:

$$p\lambda^2 + q\lambda + r = 0 \tag{3.27}$$

with:

$$p = mVJ_G \; ; \; q = m\left(C_F a^2 + C_R b^2\right) + J_G\left(C_F + C_R\right) \; ; \; r = \frac{C_F C_R L^2}{V}\left(1 + KV^2\right) \tag{3.28}$$

being K the understeer coefficient defined by (3.8). We note that in equation (3.27) coefficients p and q are positive, whereas the sign of coefficient r can be negative for an over-steering vehicle and for:

$$V > \overline{V} = K^{-\frac{1}{2}} \tag{3.29}$$

If all coefficients in (3.27) are positive, then the two roots $\lambda_{1,2}$ have negative real part, and the system is asymptotically stable. However, if r becomes negative, one of the roots has a real positive value and the vehicle becomes unstable with a divergence-type instability.

4 Stability of a railway wheelset with primary suspension.

Railway vehicles are known to suffer an oscillation consisting of a combination of lateral motion and yaw rotation, known as hunting motion. This motion arises due to the wheel-rail contact forces generated on the wheelset having conical wheel treads.

The simplest model that can be used to justify the hunting motion considers one single wheelset attached to a bogie through springs and dampers representing the primary suspension, as shown in Figure 4.1. The model only considers the motion of the wheelset in the horizontal plane and additionally assumes that the wheelset c.o.m. moves forward with constant speed V. Therefore, the wheelset model has 2 d.o.f.s, the independent coordinates being the lateral c.o.m. displacement y and the yaw rotation σ.

We perform this study under the following simplifying assumptions:

1) the wheelset moves at constant speed V in tangent track, with no traction or braking forces applied;
2) the track is considered rigid, no track irregularities (i.e. deviations from the ideal track geometry) are considered;
3) The effect of spin creep on the creep forces and the effect of the spin moment are neglected;
4) Gravitational stiffness effects are assumed to be small and are neglected.

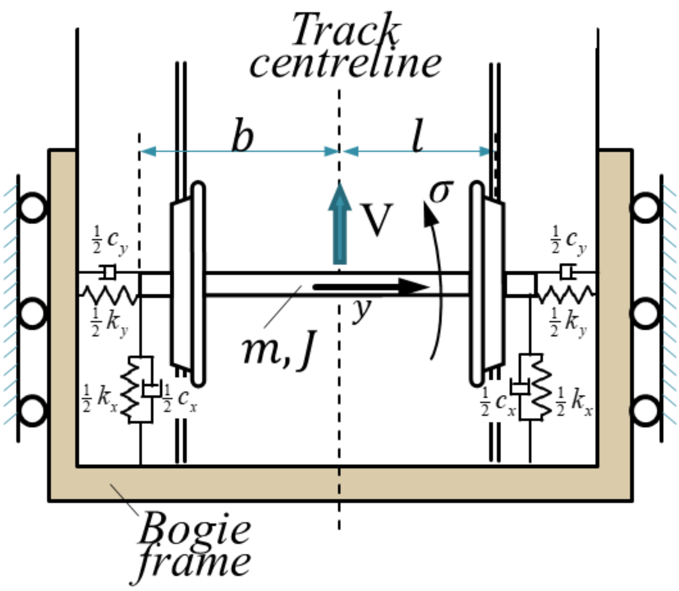


Figure 4.1: Model of a railway wheelset with primary suspension.

Figure 4.2 shows the wheelset in a displaced position and shows the forces acting on the wheelset. These are: the elastic and viscous forces generated by the primary suspension, the inertia forces and the contact force components on the left and right wheels T_{l_l}, T_{t_l}, T_{l_r}, T_{t_r}.

Assuming small displacements, the equations of motion read:

$$\begin{aligned} m\ddot{y} + c_y\dot{y} + k_y y &= -\left(T_{l_l} + T_{l_r}\right)\sin(\sigma) + \left(T_{t_l} + T_{t_r}\right)\cos(\sigma) \\ J\ddot{\sigma} + c_x b^2\dot{\sigma} + k_x b^2\sigma &= \left(T_{l_r} - T_{l_l}\right)s \end{aligned} \tag{4.1}$$

For small values of the yaw angle σ, (4.1) can be simplified to:

$$\begin{aligned} m\ddot{y} + c_y\dot{y} + k_y y &= -\left(T_{l_l} + T_{l_r}\right)\sigma + \left(T_{t_l} + T_{t_r}\right) \\ J\ddot{\sigma} + c_x b^2\dot{\sigma} + k_x b^2\sigma &= \left(T_{l_r} - T_{l_l}\right)s \end{aligned} \tag{4.1b}$$

In (4.1), the contact force components can be expressed as function of the longitudinal and lateral creepages in the left and right wheels ε_{l_l}, ε_{t_l}, ε_{l_r}, ε_{t_r}.

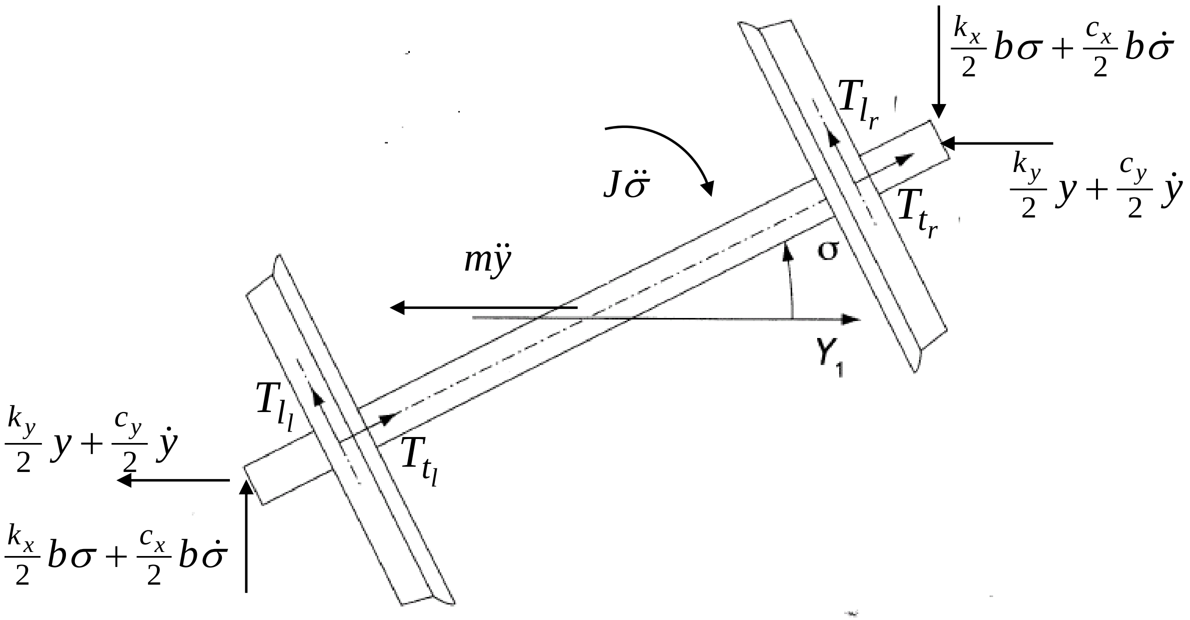


Figure 4.2: Forces acting on the wheelset.

Recalling the definition of the creepages according to (2.1), the creepages for the left and right wheels are written as:

$$\begin{aligned} \varepsilon_{l_{l,r}} &= \frac{v_{cl_{l,r}} - \omega R_{l,r}}{\left|\vec{v}_{c_{l,r}}\right|} \cong \frac{v_{cl_{l,r}} - \omega R_{l,r}}{V} \\ \varepsilon_{t_{l,r}} &= \frac{v_{ct_{l,r}}}{\left|\vec{v}_{c_{l,r}}\right|} \cong \frac{v_{ctl,r}}{V} \end{aligned} \tag{4.2}$$

with $\vec{v}_{c_{l,r}}$ the vector representing the speed of the left and right wheel centre, $v_{cl_{l,r}}$ and $v_{ct_{l,r}}$ the longitudinal and transversal components of vectors $\vec{v}_{c_{l,r}}$, ω the angular speed of the wheelset and $R_{l,r}$ the rolling radius of the left and right wheels. Note that on account of wheel conicity, the rolling radius of the two wheels is generally different and depends on the wheelset lateral displacement. If the wheels have conical shape and are resting on rails shaped as sharp edges, the dependence of the rolling radii on the wheelset lateral displacement can be written in the form:

$$\begin{aligned} R_l &= R_0 - \gamma y \\ R_r &= R_0 + \gamma y \end{aligned} \tag{4.3}$$

with γ the angle of the cone. When real wheel and rail profiles are considered, the rolling radii depend non-linearly on the lateral wheelset displacement, but for small values of the lateral displacement a linearised relationship in the form (4.3) can be obtained. In this case, γ is referred to as the *equivalent conicity* for the considered wheel-rail couple.
The longitudinal and lateral components of the speed of the two wheel centres are:

$$\begin{aligned} v_{cl_l} &= V\cos(\sigma) - \dot{y}\sin(\sigma) - s\dot{\sigma} \cong V - s\dot{\sigma} \\ v_{cl_r} &= V\cos(\sigma) - \dot{y}\sin(\sigma) + s\dot{\sigma} \cong V + s\dot{\sigma} \\ v_{ct_l} &= V\sin(\sigma) + \dot{y}\cos(\sigma) \cong V\sigma + \dot{y} \\ v_{ct_r} &= V\sin(\sigma) + \dot{y}\cos(\sigma) \cong V\sigma + \dot{y} \end{aligned} \tag{4.4}$$

Replacing (4.3) and (4.4) in (4.2):

$$\begin{aligned} \varepsilon_{l_l} &= -\varepsilon_{l_r} = -\frac{s\dot{\sigma}}{V} + \gamma\frac{\omega}{V}y = -\frac{s\dot{\sigma}}{V} + \frac{\gamma}{R_0}y \\ \varepsilon_{t_l} &= \varepsilon_{t_r} = \sigma + \frac{\dot{y}}{V} \end{aligned} \tag{4.5}$$

Starting from the general non-linear relationship (2.17b) between the creepage components and the creep forces, we derive a linearised relationship around the steady state condition:

$$\varepsilon_l = 0 \quad ; \quad \varepsilon_t = 0$$

which is valid for a wheelset moving in tangent (straight) track with no traction / braking forces applied. The linearised approximation of (2.17) reads:

$$\begin{aligned} T_l &= -\left[\mu_l(0,0,N) + \left.\frac{\partial\mu_l}{\partial\varepsilon_l}\right|_0 \varepsilon_l + \left.\frac{\partial\mu_l}{\partial\varepsilon_t}\right|_0 \varepsilon_t\right]N \\ T_t &= -\left[\mu_t(0,0,N) + \left.\frac{\partial\mu_t}{\partial\varepsilon_l}\right|_0 \varepsilon_l + \left.\frac{\partial\mu_t}{\partial\varepsilon_t}\right|_0 \varepsilon_t\right]N \end{aligned} \tag{4.6}$$

where symbol $\left.\right|_0$ denotes the quantity is evaluated for zero longitudinal and transversal creepages and for the nominal (static) value of the normal force N.
Figure 4.3 shows the trend of μ_l and μ_t with ε_l and ε_t: we note that for any value of the transversal creepage ε_t, μ_l=0 for ε_l=0. In the same way, if we neglect the effect of spin, μ_t=0 for ε_t=0 regardless the value of ε_l.
Therefore, on account of the particular steady state condition considered, we have:

$$\mu_l(0,0,N) = 0 \quad ; \quad \mu_t(0,0,N)$$

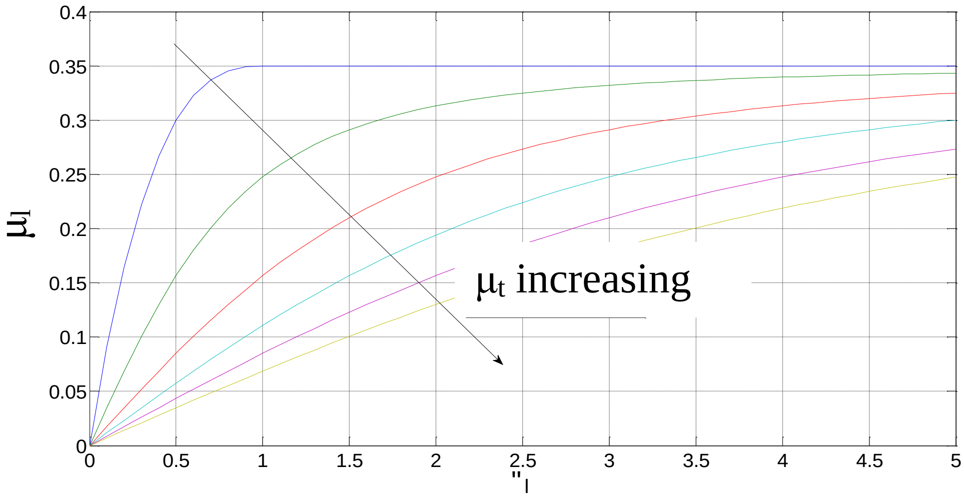
$$\left.\frac{\partial\mu_l}{\partial\varepsilon_t}\right|_0 = 0 \qquad ; \quad \left.\frac{\partial\mu_t}{\partial\varepsilon_l}\right|_0 = 0$$

According to the notation introduced by Kalker [1], we define:

$$\left.\frac{\partial \mu_l}{\partial \varepsilon_l}\right|_0 N = f_{11} \qquad ; \qquad \left.\frac{\partial \mu_t}{\partial \varepsilon_t}\right|_0 N = f_{22}$$

so that (4.6) becomes:

$$\begin{aligned} T_l &= -f_{11}\varepsilon_l \\ T_t &= -f_{22}\varepsilon_t \end{aligned} \tag{4.6b}$$



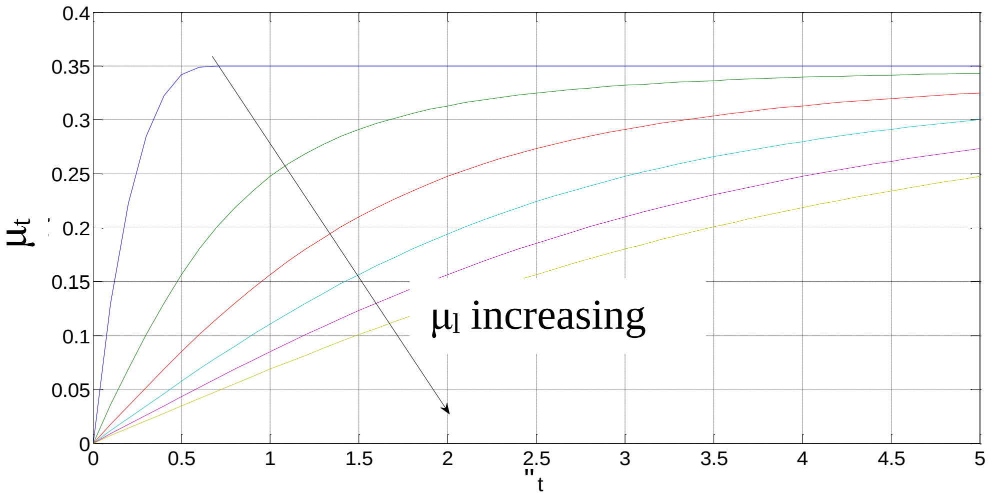


Figure 4.3: Dependence of the generalised friction coefficients μ_l and μ_t on the creepage components ε_l and ε_t.

Replacing (4.5) and (4.6b) in (4.1b) the following is obtained:

$$\begin{aligned} m\ddot{y} + c_y\dot{y} + k_y y &= -f_{22}2\left(\sigma + \frac{\dot{y}}{V}\right) \\ J\ddot{\sigma} + c_x b^2\dot{\sigma} + k_x b^2\sigma &= -f_{11}2\left(\frac{s\dot{\sigma}}{V} - \frac{\gamma}{R_0}y\right)s \end{aligned} \tag{4.7}$$

Equation (4.7) can be rewritten in matrix form as:

$$[m]\underline{\ddot{x}} + ([c_s]+[c_F])\underline{\dot{x}} + ([k_s]+[k_F])\underline{x} = \underline{0} \tag{4.8}$$

with:

$$\underline{x}=\begin{Bmatrix} y \\ \sigma \end{Bmatrix} \;;\; [m]=\begin{bmatrix} m & 0 \\ 0 & J \end{bmatrix} \;;\; [c_s]=\begin{bmatrix} c_y & 0 \\ 0 & c_x b^2 \end{bmatrix} \;;\; [k_s]=\begin{bmatrix} k_y & 0 \\ 0 & k_x b^2 \end{bmatrix} \tag{4.9}$$

and with the following expression of the damping and stiffness matrices added by the field force:

$$[c_F]=\begin{bmatrix} \dfrac{2f_{22}}{V} & 0 \\ 0 & \dfrac{2f_{11}s^2}{V} \end{bmatrix} \;;\; [k_F]=\begin{bmatrix} 0 & 2f_{22} \\ -\dfrac{2f_{11}\gamma s}{R_0} & 0 \end{bmatrix} \tag{4.10}$$

We see from (4.10) that $[k_F]$ is non-symmetric: this shows that the positional component of the force field is non-conservative. Furthermore, the diagonal terms of the total stiffness matrix $[k_T]=[k_s]+[k_F]$ have opposite sign and are large in absolute value, so that the condition for the occurrence of flutter instability:

$$\bar{K}_{Txy}\bar{K}_{Tyx}<0 \text{ and } \left|\bar{K}_{Txy}\bar{K}_{Tyx}\right|>\left(\frac{\bar{K}_{Txx}-\bar{K}_{Tyy}}{2}\right)^2 \tag{4.11}$$

introduced in the previous lesson is normally satisfied. We note however the above condition was sufficient to produce flutter instability in case no damping was applied on the system. In the case considered here, instead, damping is provided both by the structural damping matrix $[c_s]$ and by matrix $[c_F]$ from the field force. Indeed, we note that $[c_F]$ is positive definite and hence represents an equivalent viscous damping effect applied on the wheelset. Because the terms in $[c_F]$ are inversely proportional to the wheelset speed *V*, at low speed the damping introduced by the contact forces (and additionally by the structural dampers) will be large enough to dissipate the energy introduced by the positional component of the force field. With increasing speed however, energy dissipation decreases until it is no longer sufficient to dissipate the energy introduced by $[k_F]$, and flutter instability takes place.

Based on the above, we conclude that a *linear critical speed* can be defined for the wheelset as the limit speed above which the steady state motion of the wheelset in tangent track becomes unstable.

Although in this lesson we are concerned with the stability analysis of the steady-state solution for the wheelset motion, we note that based on the wheelset's non-linear equations of motion a periodic motion (limit cycle) called *hunting motion* is obtained at speeds above a threshold named the wheelset's *non-linear critical speed*. This speed is generally lower than the wheelset's linear critical speed, resulting in a bifurcation diagram that corresponds to a Hopf-type bifurcation, cf. Figure 4.3.

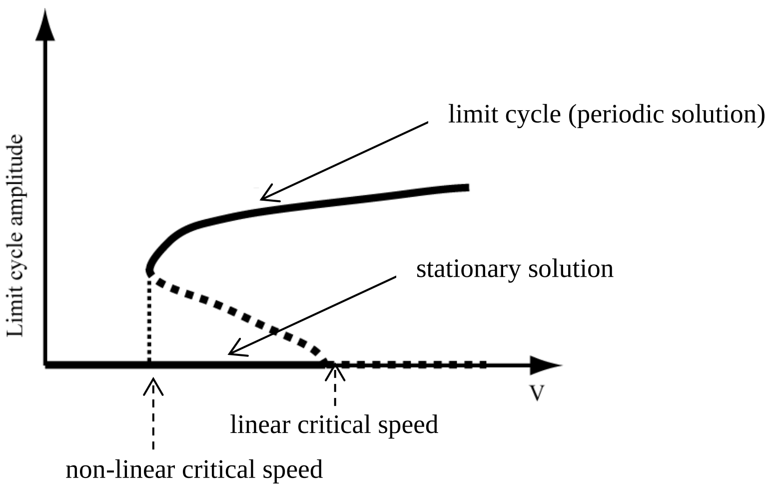


Figure 4.3: Bifurcation diagram for a railway wheelset.

Despite the wheelset limit cycle is orbitally stable [2], the vibrations associated with the lateral and yaw movement of the wheelset can be harmful to ride comfort and, in severe cases, even to running safety. It is therefore necessary to design the railway vehicle in order to avoid the occurrence of the hunting motion. Looking at the linearised equation of motion of the system, cf. equations (4.8-4.10), it is observed that:

- an increase in the primary suspension stiffness increases the diagonal terms of the total stiffness matrix $[k_T] = [k_s] + [k_F]$ thereby requiring that larger contact forces are generated before (4.11) is satisfied. Therefore, this measure is generally increasing the (linear and non-linear) critical speed of the vehicle;
- a higher wheel conicity increases one of the non-diagonal terms of the total stiffness matrix $[k_T]$ and hence lowers the (linear and non-linear) critical speed of the vehicle.

5 REFERENCES

[1] J.J. Kalker. Three-Dimensional Elastic Bodies in Rolling Contact. Solid Mechanics and its Applications. Kluwer Academic Publishers, 1990.
[2] L. Meirovitch, Elements of Vibration Analysis, McGraw-Hill Education, 1986.